

Администрирование сетевых подсистем

Лабораторная работа №14

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Цель работы

Получение практических навыков по настройке SMB-ресурсов, управлению доступом пользователей и интеграцией Samba в сетевое окружение Linux.

Настройка сервера Samba

Установка и подготовка окружения

```
Upgraded:
libldb-4.21.3-113.el10_0.x86_64
libwbclient-4.21.3-113.el10_0.x86_64
samba-common-4.21.3-113.el10_0.noarch
libsmclient-4.21.3-113.el10_0.x86_64
samba-client-libs-4.21.3-113.el10_0.x86_64
samba-common-libs-4.21.3-113.el10_0.x86_64

Installed:
cifs-utils-7.1-2.el10.x86_64
samba-4.21.3-113.el10_0.x86_64
samba-common-tools-4.21.3-113.el10_0.x86_64
samba-ldb-ldap-modules-4.21.3-113.el10_0.x86_64
libnetapi-4.21.3-113.el10_0.x86_64
samba-client-4.21.3-113.el10_0.x86_64
samba-dcerpc-4.21.3-113.el10_0.x86_64
samba-libs-4.21.3-113.el10_0.x86_64

Complete!
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# groupadd -g 1010 sambagroup
[root@server.manovichkov.net ~]# usermod -aG sambagroup manovichkov
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# mkdir -p /srv/smbashare
[root@server.manovichkov.net ~]#
```

Рис. 1: Создание группы и каталога

Конфигурация smb.conf на сервере

```
root@server:~ - sudo -i
smb.conf [----] 32 L: [ 20*30 50/ 50] *(1068/1068b) <EOF> [*][X]

-----># Install samba-usershares package for support
----->include = /etc/samba/usershares.conf

[homes]
----->comment = Home Directories
----->valid users = %S, %D%%S
----->browseable = No
----->read only = No
----->inherit acls = Yes

[printers]
----->comment = All Printers
----->path = /var/tmp
----->printable = Yes
----->create mask = 0600
----->browseable = No

[print$]
----->comment = Printer Drivers
----->path = /var/lib/samba/drivers
-----># printadmin is a local group
----->write list = printadmin root
----->force group = printadmin
----->create mask = 0664
----->directory mask = 0775

[sambashare]
----->comment = My Samba Share
----->path = /srv/sambashare
----->write list = @sambagroup

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullOn 10Quit
```

Запуск службы и проверка работы

```
● smb.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smb.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-11-01 10:57:04 UTC; 14s ago
     Invocation: e78984f5d2ad4466a86f9f8a050e63c9
       Docs: man:smbd(8)
             man:samba(7)
             man:smb.conf(5)
    Main PID: 16567 (smbd)
      Status: "smbd: ready to serve connections..."
        Tasks: 3 (limit: 10381)
       Memory: 13M (peak: 13.3M)
          CPU: 32ms
    CGroup: /system.slice/smb.service
            └─16567 /usr/sbin/smbd --foreground --no-process-group
              └─16570 /usr/sbin/smbd --foreground --no-process-group
                └─16571 /usr/sbin/smbd --foreground --no-process-group

Nov 01 10:57:04 server.manovichkov.net systemd[1]: Starting smb.service - Samba SMB Daemon...
Nov 01 10:57:04 server.manovichkov.net smbd[16567]: [2025/11/01 10:57:04.033881, 0] ../../source3/smbd/server.c:1965:
Nov 01 10:57:04 server.manovichkov.net smbd[16567]:   smbd version 4.21.3 started.
Nov 01 10:57:04 server.manovichkov.net smbd[16567]:   Copyright Andrew Tridgell and the Samba Team 1992-2024
Nov 01 10:57:04 server.manovichkov.net systemd[1]: Started smb.service - Samba SMB Daemon.
[root@server.manovichkov.net ~]# smbclient -L //server
Password for [MANOVICHKOV-NET\root]:
Anonymous login successful

      Sharename      Type      Comment
      -----
      print$         Disk      Printer Drivers
      sambashare     Disk      My Samba Share
      IPC$           IPC       IPC Service (Samba 4.21.3)

SMB1 disabled -- no workgroup available
[root@server.manovichkov.net ~]#
```

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba</short>
  <description>This option allows you to access and participate in Windows file and printer sharing networks. You need
the samba package installed for this option to be useful.</description>
  <include service="samba-client"/>
  <port protocol="tcp" port="139"/>
  <port protocol="tcp" port="445"/>
</service>
/usr/lib/firewalld/services/samba.xml (END)
```

Рис. 4: Просмотр конфигурации firewalld

Настройка прав доступа и SELinux

```
[root@server.manovichkov.net ~]# firewall-cmd --add-service=samba
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=samba --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --reload
success
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# chgrp sambagroup /srv/sambashare/
[root@server.manovichkov.net ~]# chmod g=rwx /srv/sambashare/
[root@server.manovichkov.net ~]# cd /srv
[root@server.manovichkov.net srv]# ls -Z
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:var_t:s0 sambashare
[root@server.manovichkov.net srv]# semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
[root@server.manovichkov.net srv]# restorecon -vR /srv/sambashare/
Relabeled /srv/sambashare from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:samba_share_t:s0
[root@server.manovichkov.net srv]# ls -Z
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:samba_share_t:s0 sambashare
[root@server.manovichkov.net srv]# setsebool samba_export_all_rw 1
[root@server.manovichkov.net srv]# setsebool samba_export_all_rw 1 -P
[root@server.manovichkov.net srv]#
```

Рис. 5: SELinux и настройки прав

```
[manovichkov@server.manovichkov.net ~]$ id
uid=1001(manovichkov) gid=1001(manovichkov) groups=1001(manovichkov),10(wheel) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[manovichkov@server.manovichkov.net ~]$ newgrp sambagroup
[manovichkov@server.manovichkov.net ~]$ id
uid=1001(manovichkov) gid=1010(sambagroup) groups=1010(sambagroup),10(wheel),1001(manovichkov) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[manovichkov@server.manovichkov.net ~]$ cd /srv/sambashare/
[manovichkov@server.manovichkov.net sambashare]$ touch manovichkov@server.txt
[manovichkov@server.manovichkov.net sambashare]$ ls
manovichkov@server.txt
[manovichkov@server.manovichkov.net sambashare]$
```

Рис. 6: Проверка записи в ресурс

```
[root@server.manovichkov.net srv]#  
[root@server.manovichkov.net srv]# smbpasswd -L -a manovichkov  
New SMB password:  
Retype new SMB password:  
Added user manovichkov.  
[root@server.manovichkov.net srv]#  
[root@server.manovichkov.net srv]#
```

Рис. 7: Добавление SMB-пользователя

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba Client</short>
  <description>This option allows you to access Windows file and printer sharing
networks. You need the samba-client package installed for this option to be use
ful.</description>
  <include service="netbios-ns"/>
  <port protocol="udp" port="138"/>
</service>
/usr/lib/firewalld/services/samba-client.xml (END)
```

Рис. 8: Файл samba-client.xml

Настройка firewalld на клиенте

```
[root@client.manovichkov.net ~]# less /usr/lib/firewalld/services/samba-client.xml
[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]# firewall-cmd --add-service=samba-client
success
[root@client.manovichkov.net ~]# firewall-cmd --add-service=samba-client --permanent
success
[root@client.manovichkov.net ~]# firewall-cmd --reload
success
[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]# groupadd -g 1010 sambagroup
[root@client.manovichkov.net ~]# usermod -aG sambagroup manovichkov
[root@client.manovichkov.net ~]#
```

Рис. 9: Настройка firewalld для Samba-client

Конфигурация smb.conf на клиенте

```
smb.conf      [----] 35 L:[ 1+10 11/ 46] *(346 / 979b) 0010 0x00A  [*][X]
# See smb.conf.example for a more detailed config file or
# read the smb.conf manpage.
# Run 'testparm' to verify the config is correct after
# you modified it.
#
# Note:
# SMB1 is disabled by default. This means clients without support for SMB2 or
# SMB3 are no longer able to connect to smbd (by default).

[global]
<----->workgroup = MANOVICHKOV-NET
<----->security = user

<----->passdb backend = tdbsam

<----->printing = cups
<----->printcap name = cups
<----->load printers = yes
<----->cups options = raw

<-----># Install samba-usershares package for support
<----->include = /etc/samba/usershares.conf

[homes]
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn10Quit
```

```
[ -k|--kerberos] [-v|--version] [OPTIONS] service <password>
[root@client.manovichkov.net ~]# smbclient -L //server -U manovichkov
Password for [MANOVICHKOV-NET\manovichkov]:

      Sharename      Type      Comment
      -----      -
      print$         Disk      Printer Drivers
      sambashare      Disk      My Samba Share
      IPC$           IPC       IPC Service (Samba 4.21.3)
      manovichkov     Disk      Home Directories

SMB1 disabled -- no workgroup available
[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]# mkdir /mnt/samba
[root@client.manovichkov.net ~]# mount -o username=manovichkov,user,rw
```

Рис. 11: smbclient под manovichkov

```
root@client.manovichkov.net ~]# mount | grep mnt
server.manovichkov.net:/srv/nfs on /mnt/nfs type nfs4 (rw,relatime,vers=4.2,rsiz
e=262144,wsiz=262144,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clie
ntaddr=192.168.1.30,local_lock=none,addr=192.168.1.1,_netdev)
//server/sambashare on /mnt/samba type cifs (rw,nosuid,nodev,noexec,relatime,ver
s=3.1.1,cache=strict,upcall_target=app,username=manovichkov,uid=1001,forceuid,gi
d=1010,forcegid,addr=192.168.1.1,file_mode=0755,dir_mode=0755,soft,nounix,server
no,mapposix,reparsing=nfs,rsiz=4194304,wsiz=4194304,bsiz=1048576,retrans=1,ech
o_interval=60,actimeo=1,closetimeo=1,user)
root@client.manovichkov.net ~]# █
```

Рис. 12: Монтирование SMB-ресурса

Создание файла на смонтированном ресурсе

```
[manovichkov@client.manovichkov.net samba]$ newgrp sambagroup
[manovichkov@client.manovichkov.net samba]$ id
uid=1001(manovichkov) gid=1010(sambagroup) groups=1010(sambagroup),10(wheel),100
1(manovichkov) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[manovichkov@client.manovichkov.net samba]$
[manovichkov@client.manovichkov.net samba]$ ls
manovichkov@server.txt
[manovichkov@client.manovichkov.net samba]$ touch manovichkov@client.txt
[manovichkov@client.manovichkov.net samba]$ ls
manovichkov@client.txt manovichkov@server.txt
[manovichkov@client.manovichkov.net samba]$
```

Рис. 13: Создание файла на клиенте

Вывод

В ходе лабораторной работы был настроен сервер Samba, обеспечивающий общий доступ к данным в локальной сети. Проверена работа прав доступа, SELinux, firewall, смонтирован сетевой ресурс на клиенте, а также подготовлены автоматизационные скрипты для Vagrant.